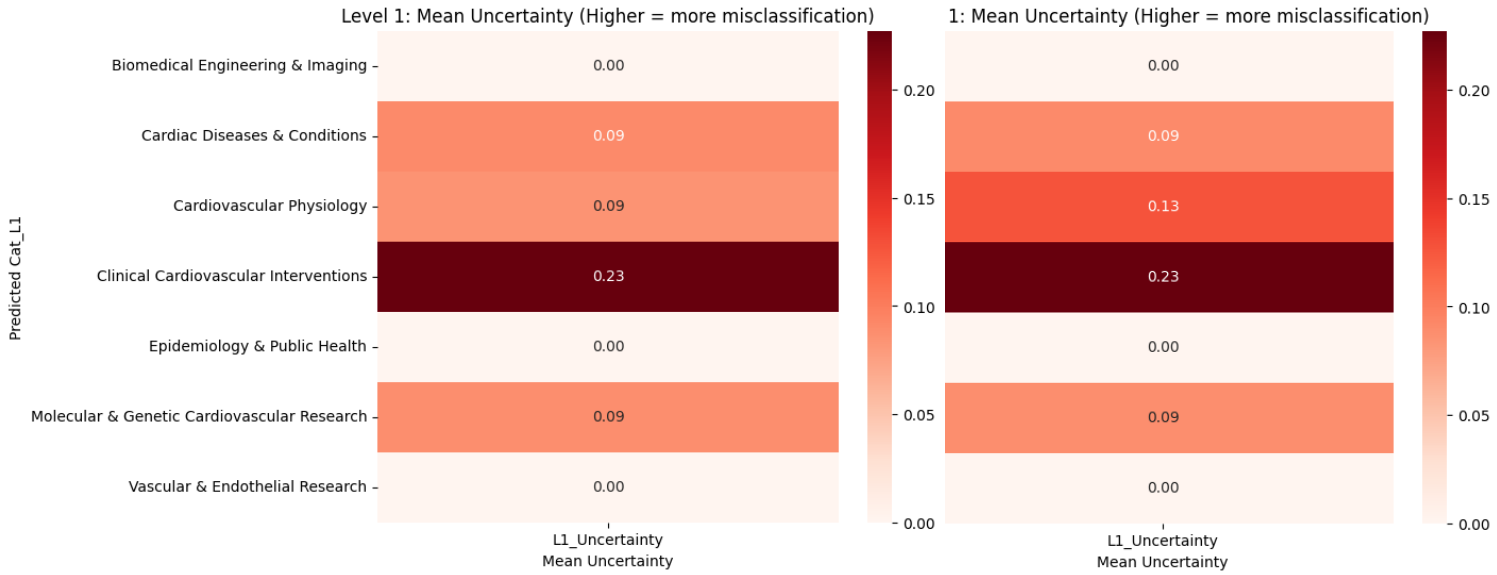
Inconsistency Rates for GPT 4.5 (200 Rows) w/ GPT 3.5-Turbo (8x23x64 labels)

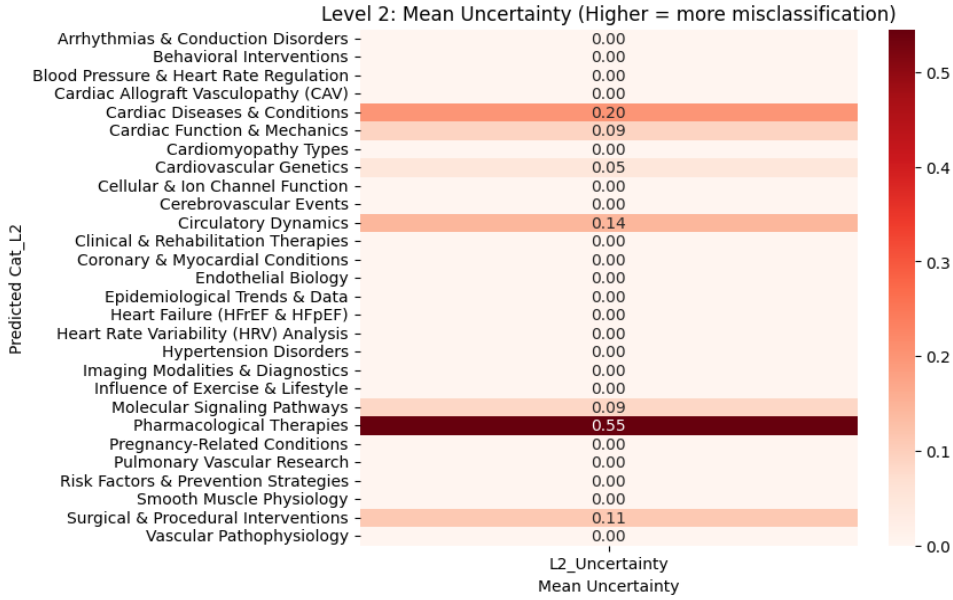
Run	Cat_L1 Inconsistency (%)	Cat_L2 Inconsistency (%)	Cat_L3 Inconsistency (%)
1	21 (10.50%)	19 (9.50%)	76 (38.00%)
2	19 (9.50%)	17 (8.50%)	83 (41.50%)
Average	**20 (10.00%)**	**18 (9.00%)**	**79.5 (39.75%)**

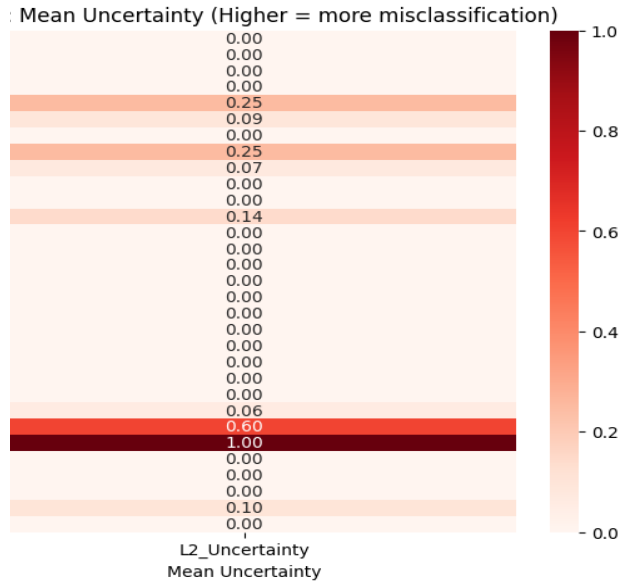
Inconsistency Rates for GPT o1 (200 Rows) w/ GPT 3.5-Turbo (7x22x44 labels)

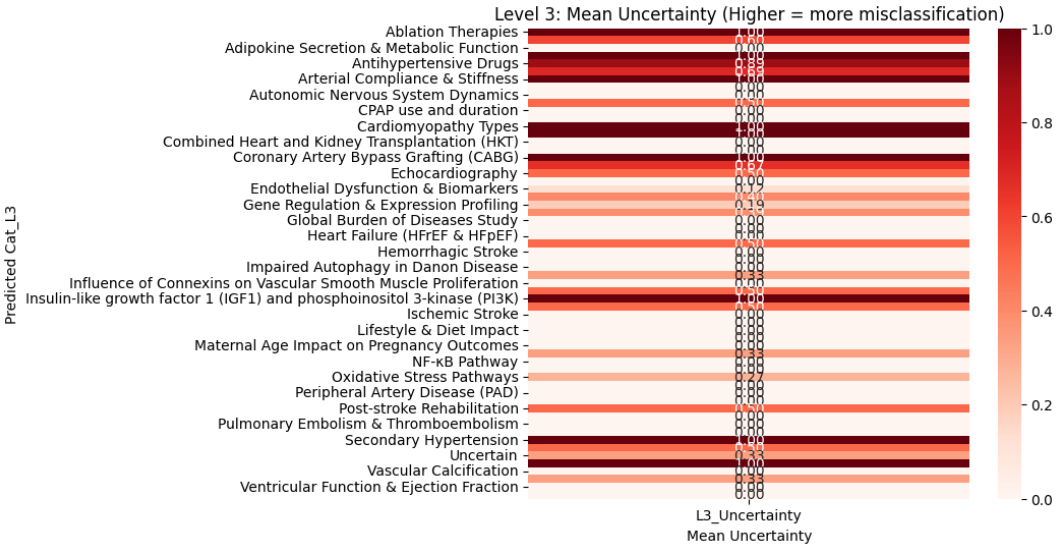
Run	Cat_L1 Inconsistency (%)	Cat_L2 Inconsistency (%)	Cat_L3 Inconsistency (%)
1	5 (2.50%)	13 (6.50%)	82 (41.00%)
2	5 (2.50%)	13 (6.50%)	80 (40.00%)
Average	**5 (2.50%)**	**13 (6.50%)**	**81 (40.50%)**

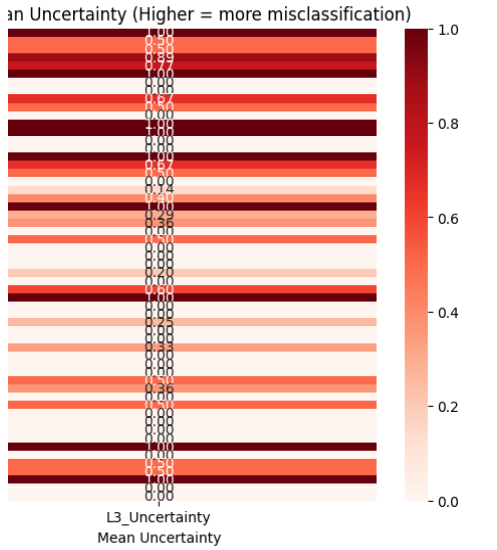
<u>GPT 4.5 Heatmaps Run#1 vs Run#2:</u>



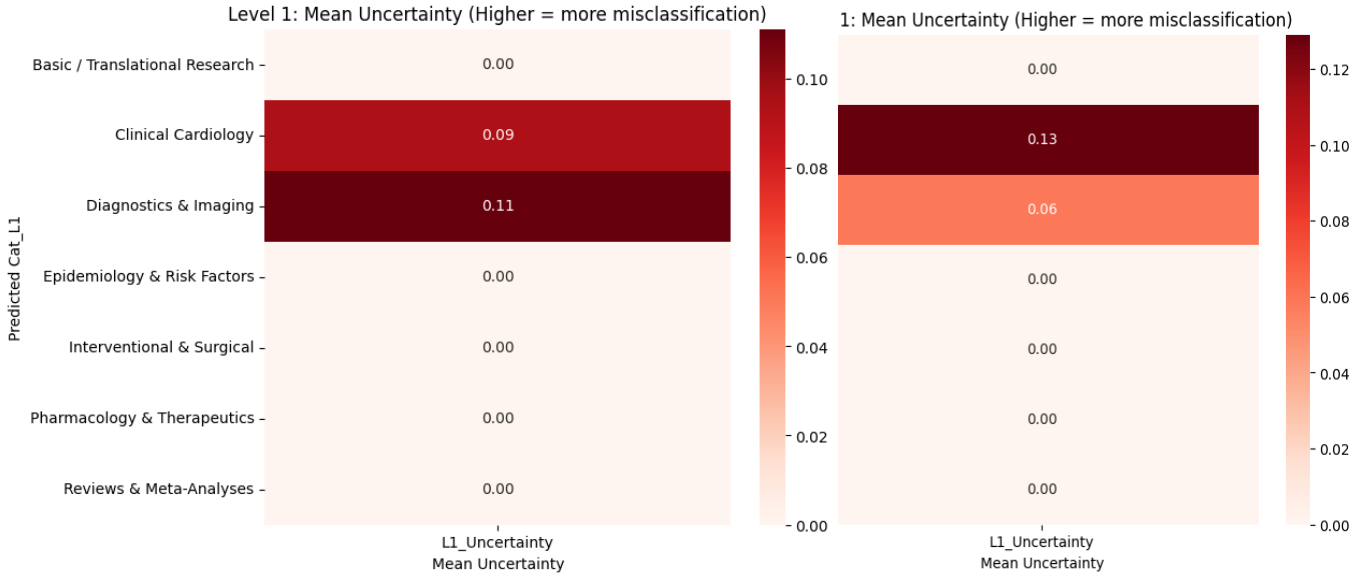

Level 2: Mean Uncertainty (Higher = more misclassification)
Predicted Cat_L2
Arrhythmias & Conduction Disorders 0.00
Behavioral Interventions 0.00
Blood Pressure & Heart Rate Regulation 0.00
Cardiac Allograft Vasculopathy (CAV) 0.00
Cardiac Diseases & Conditions 0.20
Cardiac Function & Mechanics 0.09
Cardiomyopathy Types 0.00
Cardiovascular Genetics 0.05
Cellular & Ion Channel Function 0.00
Cerebrovascular Events 0.00
Circulatory Dynamics 0.14
Clinical & Rehabilitation Therapies 0.00
Coronary & Myocardial Conditions 0.00
Endothelial Biology 0.00
Epidemiological Trends & Data 0.00
Heart Failure (HFrEF & HFpEF) 0.00
Heart Rate Variability (HRV) Analysis 0.00
Hypertension Disorders 0.00
Imaging Modalities & Diagnostics 0.00
Influence of Exercise & Lifestyle 0.00
Molecular Signaling Pathways 0.09
Pharmacological Therapies 0.55
Pregnancy-Related Conditions 0.00
Pulmonary Vascular Research 0.00
Risk Factors & Prevention Strategies 0.00
Smooth Muscle Physiology 0.00
Surgical & Procedural Interventions 0.11
Vascular Pathophysiology 0.00
L2_Uncertainty
Mean Uncertainty
0.0
0.1
0.2
0.3
0.4
0.5

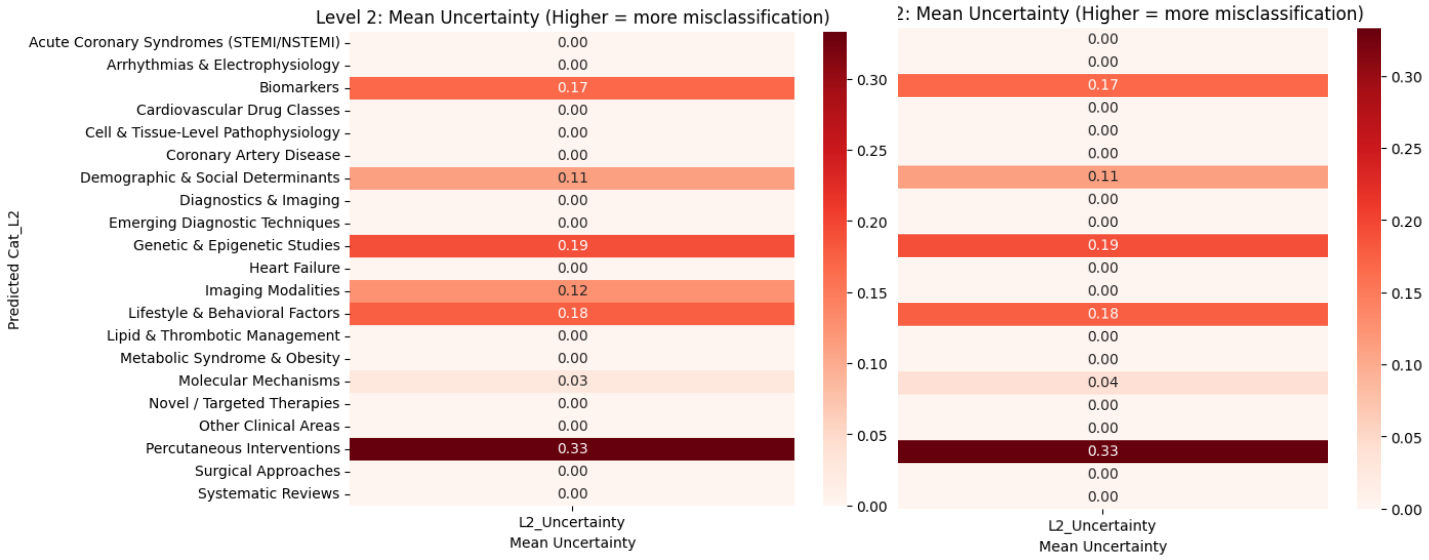

Mean Uncertainty (Higher = more misclassification)
0.00
0.00
0.00
0.00
0.25
0.09
0.00
0.25
0.07
0.00
0.00
0.14
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.06
0.60
1.00
0.00
0.00
0.00
0.10
0.00
L2_Uncertainty
Mean Uncertainty
0.0
0.2
0.4
0.6
0.8
1.0

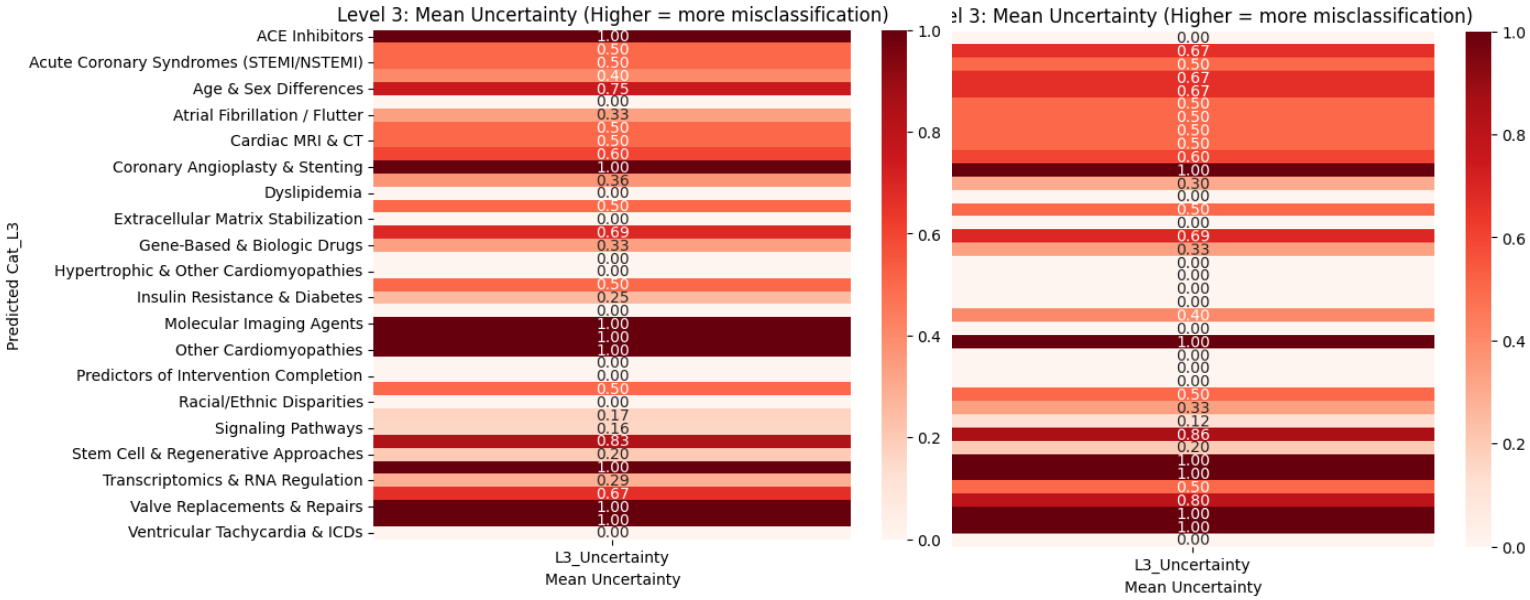

Level 3: Mean Uncertainty (Higher = more misclassification)
Predicted Cat_L3
Ablation Therapies 1.00
Adipokine Secretion & Metabolic Function 0.50
Antihypertensive Drugs 0.00
Arterial Compliance & Stiffness 0.89
Autonomic Nervous System Dynamics 1.00
CPAP use and duration 0.00
Cardiomyopathy Types 0.50
Combined Heart and Kidney Transplantation (HKT) 1.00
Coronary Artery Bypass Grafting (CABG) 0.00
Echocardiography 1.00
Endothelial Dysfunction & Biomarkers 0.57
Gene Regulation & Expression Profiling 0.00
Global Burden of Diseases Study 0.19
Heart Failure (HFrEF & HFpEF) 0.25
Hemorrhagic Stroke 0.00
Impaired Autophagy in Danon Disease 0.00
Influence of Connexins on Vascular Smooth Muscle Proliferation 0.00
Insulin-like growth factor 1 (IGF1) and phosphoinositol 3-kinase (PI3K) 0.50
Ischemic Stroke 1.00
Lifestyle & Diet Impact 0.50
Maternal Age Impact on Pregnancy Outcomes 0.00
NF-κB Pathway 0.00
Oxidative Stress Pathways 0.00
Peripheral Artery Disease (PAD) 0.33
Post-stroke Rehabilitation 0.00
Pulmonary Embolism & Thromboembolism 0.50
Secondary Hypertension 0.00
Uncertain 0.00
Vascular Calcification 1.00
Ventricular Function & Ejection Fraction 0.33
0.00
L3_Uncertainty
Mean Uncertainty
0.0
0.2
0.4
0.6
0.8
1.0


an Uncertainty (Higher = more misclassification)
1.00
0.50
0.89
0.77
1.00
0.00
0.57
0.50
0.00
1.00
0.00
1.00
0.57
0.00
0.14
0.50
1.00
0.29
0.36
0.00
0.50
0.00
0.00
0.20
0.00
0.50
1.00
0.00
0.25
0.00
0.33
0.00
0.00
0.50
0.36
0.00
0.50
0.00
0.00
0.00
1.00
0.00
0.50
0.50
1.00
0.00
L3_Uncertainty
Mean Uncertainty
0.0
0.2
0.4
0.6
0.8
1.0

GPT o1 Heatmaps Run#1 vs Run#2:







<u>Hierarchical Categorization Labels:</u>

O1 (7x22x44):

1. Basic / Translational Research

  1A. Molecular Mechanisms

    1A.i. Signaling Pathways

    1A.ii. Receptors & Ion Channels

  1B. Genetic & Epigenetic Studies

    1B.i. Transcriptomics & RNA Regulation

    1B.ii. Gene Editing & Knockouts

  1C. Cell & Tissue-Level Pathophysiology

    1C.i. Cardiomyocyte Hypertrophy & Remodeling

    1C.ii. Stem Cell & Regenerative Approaches

2. Clinical Cardiology

2A. Coronary Artery Disease

2A.i. Angina & Chronic Ischemic Syndromes

2A.ii. Acute Coronary Syndromes (STEMI/NSTEMI)

2B. Heart Failure

2B.i. Systolic vs. Diastolic Dysfunction

2B.ii. Advanced Therapies (LVAD, Transplant)

2C. Arrhythmias & Electrophysiology

2C.i. Atrial Fibrillation / Flutter

2C.ii. Ventricular Tachycardia & ICDs

2D. Other Clinical Areas

2D.i. Valvular Heart Disease

2D.ii. Hypertrophic & Other Cardiomyopathies

3. Epidemiology & Risk Factors

3A. Lifestyle & Behavioral Factors

3A.i. Smoking, Tobacco, & Vaping

3A.ii. Diet & Physical Activity

3B. Metabolic Syndrome & Obesity

3B.i. Insulin Resistance & Diabetes

3B.ii. Dyslipidemia

3C. Demographic & Social Determinants

3C.i. Age & Sex Differences

3C.ii. Racial/Ethnic Disparities

4. Diagnostics & Imaging

- 4A. Biomarkers
  - 4A.i. Troponins, Natriuretic Peptides
  - 4A.ii. Inflammatory Markers (CRP, IL-6, etc.)
- 4B. Imaging Modalities
  - 4B.i. Echocardiography & Ultrasound
  - 4B.ii. Cardiac MRI & CT
- 4C. Emerging Diagnostic Techniques
  - 4C.i. AI & Machine Learning Tools
  - 4C.ii. Molecular Imaging Agents

5. Interventional & Surgical
- 5A. Percutaneous Interventions
  - 5A.i. Coronary Angioplasty & Stenting
  - 5A.ii. Transcatheter Valve Procedures (TAVR)
- 5B. Surgical Approaches
  - 5B.i. Coronary Artery Bypass Graft (CABG)
  - 5B.ii. Valve Replacements & Repairs
- 5C. Advanced Cardiac Devices
  - 5C.i. Pacemakers & Defibrillators (ICDs)
  - 5C.ii. Ventricular Assist Devices

6. Pharmacology & Therapeutics
- 6A. Cardiovascular Drug Classes
  - 6A.i. Beta-Blockers, ACE Inhibitors, ARBs
  - 6A.ii. Antiarrhythmics & Calcium Channel Blockers
- 6B. Lipid & Thrombotic Management

6B.i. Statins & PCSK9 Inhibitors

6B.ii. Antiplatelets & Anticoagulants

6C. Novel / Targeted Therapies

6C.i. Gene-Based & Biologic Drugs

6C.ii. Immunomodulatory & Anti-Inflammatory Agents

7. Reviews & Meta-Analyses

7A. Systematic Reviews

7A.i. RCT-Based Summaries

7A.ii. Observational Summaries

7B. Narrative & Thematic Reviews

7B.i. Mechanistic Focus

7B.ii. Clinical Focus

7C. Guidelines & Consensus

7C.i. Professional Society Guidelines

7C.ii. Expert Consensus Documents

4.5 (8x23x64):

I. Cardiovascular Physiology

A. Blood Pressure & Heart Rate Regulation

1. Baroreflex Mechanisms & Regulation
2. Heart Rate Variability (HRV) Analysis
3. Influence of Exercise & Lifestyle
4. Autonomic Nervous System Dynamics

B. Cardiac Function & Mechanics

1. Ventricular Function & Ejection Fraction
2. Diastolic Dysfunction
3. Cardiac Remodeling & Adaptation
4. Smooth Muscle Physiology

C. Circulatory Dynamics

1. Arterial Compliance & Stiffness
2. Hemodynamics & Flow Dynamics
3. Microcirculation & Capillary Dynamics

II. Cardiac Diseases & Conditions

A. Coronary & Myocardial Conditions

1. Acute Myocardial Infarction
2. Chronic Coronary Artery Disease
3. Myocarditis & Pericarditis
4. Heart Failure (HFrEF & HFpEF)
5. Cardiomyopathy Types

B. Arrhythmias & Conduction Disorders

1. Atrial Fibrillation & Flutter
2. Ventricular Arrhythmias
3. Bradycardia & Conduction Block
4. Pacemaker & Defibrillator Therapy

C. Hypertension Disorders

1. Essential Hypertension
2. Resistant Hypertension
3. Pulmonary Hypertension
4. Secondary Hypertension

D. Cerebrovascular Events

1. Ischemic Stroke
2. Hemorrhagic Stroke
3. Transient Ischemic Attack (TIA)
4. Post-stroke Rehabilitation

III. Molecular & Genetic Cardiovascular Research

A. Cardiovascular Genetics

1. Genetic Polymorphisms & Risk Factors
2. Gene Regulation & Expression Profiling
3. Epigenetic Modifications

B. Molecular Signaling Pathways

1. Inflammatory Cytokines
2. Oxidative Stress Pathways
3. Apoptosis & Cellular Death Pathways

C. Cellular & Ion Channel Function

1. Ion Channels (Na^+, K^+, Ca^{2+})
2. Excitation-Contraction Coupling
3. Endothelial Cell Function & Dysfunction

IV. Vascular & Endothelial Research

A. Endothelial Biology

1. Endothelial Dysfunction & Biomarkers
2. Nitric Oxide Pathways
3. Endothelial Repair & Angiogenesis

B. Vascular Pathophysiology

1. Atherosclerosis & Plaque Formation
2. Vascular Calcification
3. Peripheral Artery Disease (PAD)

C. Pulmonary Vascular Research

1. Pulmonary Embolism & Thromboembolism
2. Pulmonary Arterial Hypertension (PAH)

V. Biomedical Engineering & Imaging

A. Imaging Modalities & Diagnostics

1. Echocardiography
2. Magnetic Resonance Imaging (MRI)
3. Computed Tomography (CT)
4. Nuclear Cardiology & PET Scanning

B. Computational Cardiology

1. AI & Machine Learning in Cardiology
2. Computational Hemodynamics
3. Cardiac Image Processing & Segmentation

C. Wearable & Remote Monitoring Technology

1. ECG & Rhythm Monitoring Devices
2. Mobile Health (mHealth) Solutions
3. Telemedicine & Virtual Cardiology Care

VI. Clinical Cardiovascular Interventions

A. Pharmacological Therapies

1. Antihypertensive Drugs
2. Anticoagulants & Antiplatelets

3. Lipid-Lowering Therapies
4. Antiarrhythmic Medications

B. Surgical & Procedural Interventions
1. Coronary Artery Bypass Grafting (CABG)
2. Percutaneous Coronary Intervention (PCI)
3. Valve Repair & Replacement
4. Ablation Therapies

VII. Epidemiology & Public Health

A. Risk Factors & Prevention Strategies
1. Lifestyle & Diet Impact
2. Environmental & Social Determinants
3. Epidemiological Trends & Data

B. Health Systems & Policy
1. Healthcare Access & Outcomes
2. Economic Impact of Cardiovascular Diseases
3. Public Health Initiatives & Policies

VIII. Large Language Models & Methodological Approaches

A. NLP & Fact Extraction Techniques
1. Zero-shot & Few-shot Extraction
2. Prompt Engineering Methods

B. Similarity & Contradiction Analysis
1. Cosine Similarity Metrics
2. Semantic Embedding Techniques
3. Contradiction Detection Models

C. Topic Modeling & Textual Analysis

1. Latent Dirichlet Allocation (LDA)
2. Transformer-based Embeddings
3. Temporal & Trend Analysis